What it takes to bring open source into our products

How Ciena evaluates open source software and hardware components

Open Source at Ciena

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About me



Jamie Sanderson

Director, Software Engineering Systems

Industry engagement

Close involvement with ASIC and FPGA teams; led FPGA technology roadmap reviews with AMD/Xilinx and Intel/Altera at Ericsson and Ciena.

1995 — **Bell Northern Research**
Intern student

1997 — **Nortel**
FPGA developer, then manager of the FPGA & Algorithms team

2010 — **Ericsson**
Wireless team moved to Ericsson; continued leading FPGA & Algorithms

2016 — **Ericsson**
Managed embedded software

2017 — **Ciena**
Joined Ciena, back to managing FPGA

2020 — **Ciena**
Director, Software (embedded & DevOps); ~2 years as Director, FPGA

Today — **Ciena**
Director in software, increasingly focused on AI in software development

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Open source powers modern networks, responsibly

Open source software is in nearly every modern product, and open source hardware is emerging. Ciena embraces open source software with clear criteria, and the same criteria apply to hardware.

Accelerates innovation

Proven community-built components let our engineers focus on what makes Ciena solutions unique.

Carries shared responsibility

When open source ships in our products, as code or in silicon, its security, licensing and quality become our commitment to customers.

Demands consistent criteria

A clear, repeatable evaluation lets teams adopt open source quickly and with confidence.

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Eight criteria every component must meet

License Compliance

1

Obligations understood and met

Security & Vulnerability Management

2

Safe to ship and to keep patching

Community Health

3

Active, well-governed project

Lifecycle & Maintainability

4

Supportable for the product's life

Technical Fit & Quality

5

Right tool, built well

Supply Chain Integrity

6

Verifiable provenance end to end

Export & Regulatory Compliance

7

Shippable to every market

Intellectual Property Protection

8

Rights protected on all sides

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License Compliance

CRITERION 1 OF 8 Can we meet every obligation the license imposes?

Why it matters

Licenses define what we may do with the code or design and what we must give back. Misreading them puts customers and products at risk.

What we require

- License on Ciena's approved list (OSI for software; Solderpad, CERN-OHL for hardware)
- Copyleft terms (GPL, LGPL, AGPL, CERN-OHL-S) reviewed for distribution and linking
- Attribution and notice obligations captured in product documentation
- Compatible with the licenses of every other component in the product

Evidence we look for: SPDX license identifiers, license scan results for source and RTL, legal review record

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Security & Vulnerability Management

CRITERION 2 OF 8 Is it safe to ship, and to keep shipping?

Why it matters

Our products carry critical network traffic. Every component, in software or silicon, inherits our security commitments to customers.

What we require

- No unremediated critical or high-severity CVEs at intake
- Maintainers follow a responsive security and disclosure process
- Tracked in the product SBOM and continuous vulnerability scanning
- Fixes can be applied quickly, upstream or in-house; hardware errata have a mitigation path

Evidence we look for: CVE and errata history, OpenSSF Scorecard results, software composition analysis reports

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Community Health

CRITERION 3 OF 8 Will the project still be thriving in five years?

Why it matters

An active community delivers fixes, features and security patches. An abandoned project becomes our burden alone.

What we require

- Multiple active maintainers, ideally from different organizations
- Regular releases and recent commit activity
- Transparent governance or neutral foundation backing
- Issues and pull requests addressed in a reasonable time

Evidence we look for: Contributor counts, release cadence, foundation membership and governance documents

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Lifecycle & Maintainability

CRITERION 4 OF 8 Can we support it as long as our customers need us to?

Why it matters

Ciena products stay in service for many years, and hardware is hard to change once deployed. Every component must be supportable for the full product lifecycle.

What we require

- Supported version with a published roadmap or end-of-life policy
- A named internal owner accountable for updates
- Clear upgrade path without major rework
- Minimal local modifications: fix upstream wherever possible

Evidence we look for: Version support policy, internal ownership record, upgrade plan

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Technical Fit & Quality

CRITERION 5 OF 8 Is it the right tool, and is it built well?

Why it matters

Open source should accelerate engineering, not add complexity or put product performance and reliability at risk.

What we require

- Meets functional, performance and scalability requirements, plus timing, area and power for hardware
- Automated tests and a healthy defect trend; for hardware, a verification suite with coverage
- Compatible with our platforms, toolchains and architecture, including FPGA and EDA flows
- Justified against already-approved components to avoid duplication

Evidence we look for: Architecture review, benchmark results, test and verification coverage data

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Supply Chain Integrity

CRITERION 6 OF 8 Do we know exactly where this code came from?

Why it matters

Attacks increasingly target the software and hardware supply chain. Provenance must be verifiable from source or RTL to shipped product.

What we require

- Obtained only from official, trusted sources
- Signed releases or verifiable checksums
- Transitive dependencies identified and assessed
- Built or synthesized within Ciena's controlled, reproducible pipeline

Evidence we look for: SBOM and hardware BOM (SPDX or CycloneDX), signature verification, SLSA-aligned build records

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Export & Regulatory Compliance

CRITERION 7 OF 8 Can we ship it everywhere our customers are?

Why it matters

Cryptography, processor IP and other technologies carry export controls and regulatory requirements across global markets.

What we require

- Encryption functionality identified and classified, in software and hardware
- Export control classification reviewed before release
- Meets applicable industry and government requirements
- Country-specific restrictions documented

Evidence we look for: Export classification record, cryptography inventory, compliance sign-off

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Intellectual Property Protection

CRITERION 8 OF 8 Does it respect both Ciena's IP and the rights of others?

Why it matters

Combining open source with proprietary code or designs must never expose Ciena innovation or infringe on the rights of others.

What we require

- Clear copyright and provenance: no code of unknown origin
- No known patent encumbrances or assertions, in code or hardware designs
- Proprietary code and IP blocks kept separate where licenses require it
- Contributions back follow Ciena's approved contribution process

Evidence we look for: Provenance scan, IP review, contribution approvals

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Applying the criteria to open source hardware

EXAMPLE OpenHW Foundation CORE-V CVA5, a 32-bit RISC-V processor core for FPGAs

What's different for hardware

Hardware IP like CVA5 is SystemVerilog RTL that becomes part of an FPGA image or chip. Licenses differ from software, defects are harder to fix once deployed, and verification evidence matters as much as code quality.

How CVA5 measures up

- License: Solderpad 2.1, an Apache 2.0 extension; check notice and patent terms
- Community: OpenHW Foundation (Eclipse Foundation); derived from SFU's Taiga project
- Technical fit: configurable RV32IMAFD SystemVerilog core; check FPGA timing and area
- Lifecycle: check release cadence, maintainer activity and verification maturity

Evidence we look for: License headers in RTL, verification and coverage reports, FPGA synthesis results, export classification

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How a component earns its place in a Ciena product

Our review flow applies all eight criteria consistently, from request to retirement.



Principle: approval is not a one-time event. Every component is re-assessed as new versions, vulnerabilities and obligations emerge.

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Open source, adopted with confidence

Clear criteria let Ciena innovate faster while keeping our commitments to customers.



Legally sound

Licenses, export rules and IP obligations understood before code or silicon ships.



Secure by design

Vulnerabilities and supply chain risks managed for the life of the product.



Sustainable

Healthy communities and named owners keep every component supported.

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